

2024, Mar. 2024 Release LOC:Acute Pediatric

Bone Marrow Transplant/Stem Cell Transplant (BMT/SCT)

Overview

Select Day

Pre-procedure Day 1-7

Procedure Day

Post-procedure Day 1-14

Post-procedure Day 15-30

Post-procedure Day 31

Discharge Screens ⁽³⁷⁾

Notes

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Overview

Instruction: This subset covers the pre-procedure conditioning, infusion and post-procedure course of stem cell and bone marrow transplant recipients. Patient who experience graft failure requiring re-transplant will require a repeat round of pre-procedure conditioning before undergoing a repeat stem cell or bone marrow transplant. To review solid organ transplants the reviewer should refer to the General Surgical subset.

Introduction:

Bone marrow (BM) and stem cell transplants (SCT) are a potentially curative therapy for many hematologic disorders (Majhail et al., Biol Blood Marrow Transplant 2015, 21: 1863-9). More than 60,000 hematopoietic transplants are performed worldwide annually to treat a variety of both malignant and non-malignant conditions. Patients are treated at a transplant center and usually require a 3-4 month stay before they can return home (Hashmi, Prim Care 2016, 43: 693-701).

More than 70% of the transplants performed in the United States are peripheral blood stem cell (PBSC) transplants. Bone marrow transplants account for the remainder.

Bone marrow and stem cell transplants are currently performed for both malignant and non-malignant conditions. Some of the indications for transplant may include leukemias, lymphomas, myelodysplastic syndromes, myeloproliferative neoplasms, multiple myeloma, solid tumors (e.g., germ cell tumors, neuroblastoma, Ewings sarcoma, medulloblastoma), and non-malignant disorders (e.g., immune deficiencies, inherited metabolic disorders, hemoglobinopathies, hemophagocytic lymphohistiocytosis, severe aplastic anemia, and other marrow failure syndromes) (Majhail et al., Biol Blood Marrow Transplant 2015, 21: 1863-9; Hashmi, Prim Care 2016, 43: 693-701). There are 3 types of transplants: syngeneic (from an identical twin), allogeneic (from a related or unrelated donor), and autologous (from the patient). Donor cells can originate from the bone marrow (BM), peripheral blood stem cells (PBSC), or umbilical cord blood (UCB) (Hashmi, Prim Care 2016, 43: 693-701).

Graft failure is a condition where engraftment fails to succeed and the patient must undergo a new transplant. This occurs in < 5% of allogeneic transplant patients and is even less likely in autologous transplants (Hashmi, Prim Care 2016, 43: 693-701).

The most common complication following allogeneic transplant is graft versus host disease (GVHD). Acute graft vs host disease (GVHD) is generally seen in 3 main sites: the gastrointestinal tract, the skin, and the liver (Hashmi, Prim Care 2016, 43: 693-701). Infection is also a common complication due to a depleted immune system.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data

sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Symptom or finding within 24h)

(Excludes PO medications unless noted)

Select Day, One:● **Pre-procedure Day 1-7, One:**● **ACUTE, One:** ⁽¹⁾

- Pre-bone marrow or stem cell transplant and, **One:** ^(2, 3)
 - Inpatient submyeloablative therapy ⁽⁴⁾
 - Inpatient myeloablative therapy

● **Procedure Day, One:**● **INTERMEDIATE, All:** ⁽⁵⁾

- Administration of transplanted cells
- Isolation
- Hydration or nutrition, ≥ **One:**
 - IV fluid ^(6, 7)
 - Advancing diet as tolerated or nutritional route established ⁽⁸⁾

● **Post-procedure Day 1-14, One:**● **INTERMEDIATE, All:**

- Post bone marrow or stem cell transplant
- Awaiting engraftment
- Isolation
- Hydration or nutrition, ≥ **One:**
 - IV fluid ^(6, 7)
 - Advancing diet as tolerated or nutritional route established ⁽⁸⁾

● **CRITICAL, ≥ One:** ⁽⁹⁾● Hemodynamic instability, **Both:**● Finding, ≥ **Two:**● Heart rate, **One:** ⁽¹⁰⁾

- > 180/min, sustained (age ≤ 12 mos) ⁽¹¹⁾
- > 160/min, sustained (age > 12 mos to 4 yrs) ⁽¹¹⁾
- > 150/min, sustained (age > 4 yrs to 8 yrs) ⁽¹¹⁾
- > 140/min, sustained (age > 8 yrs to 15 yrs) ⁽¹¹⁾
- > 130/min, sustained (age > 15 yrs to < 18 yrs) ⁽¹¹⁾

● Systolic BP, **One:**

- < 70 mmHg (age ≤ 12 mos)
- < 80 mmHg (age > 1 to < 5 yrs)
- < 90 mmHg (age 5 to < 18 yrs)
- Labile ⁽¹²⁾

● MAP, **One:** ⁽¹³⁾

- < 55 mmHg (≤ 12 mos)
- < 65 mmHg (> 1 to < 18 yrs)

● Intervention, ≥ **One:**

- Blood product transfusion ⁽¹⁴⁾
- IV fluid resuscitation ≤ 2d ⁽¹⁵⁾
- Hemodynamic monitoring, invasive ⁽¹⁶⁾
- Mechanical ventilation
- NIPPV and, ≥ **One:** ⁽¹⁷⁾
 - FiO₂ ≥ 40%(0.40)
 - Respiratory intervention every 1-2h ⁽¹⁸⁾
- Oxygen ≥ 40%(0.40) ⁽¹⁹⁾
- Vasopressor, vasodilator, or inotrope continuous (excludes low or fixed dose for end stage heart failure)

● **Post-procedure Day 15-30, One:**● **ACUTE, One:**

- **Responder**, discharge expected today if clinically stable last 24h, **All:** ⁽²⁰⁾
 - Afebrile
 - ANC $\geq 500/\text{cu.mm}(500 \times 10^6/\text{L})$
 - Platelet count $> 20,000/\text{cu.mm}(20 \times 10^9/\text{L})$
 - Tolerating PO or nutritional route established ^(8, 21)
- **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽²²⁾
 - **Complication or comorbidity and clinically stable for discharge, $\geq$ One:**
 - Pain controlled or manageable ⁽²³⁾
 - GVHD resolving or stabilized
- **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - **Bone marrow or stem cell transplant and, All:**
 - Post engraftment
 - **Finding, $\geq$ One:**
 - ANC $< 500/\text{cu.mm}(500 \times 10^6/\text{L})$
 - Platelet count $\leq 20,000/\text{cu.mm}(20 \times 10^9/\text{L})$
 - **Hydration or nutrition, $\geq$ One:**
 - IV fluid ^(6, 7)
 - Advancing diet as tolerated or nutritional route established ⁽⁸⁾
- **GVHD and, Both:** ⁽²⁴⁾
 - Immunosuppressant ^(25, 26)
 - **Intervention, $\geq$ One:**
 - Initiation of new medication $\leq 2\text{d}$
 - Conversion from IV to PO $\leq 2\text{d}$ ⁽²⁷⁾
 - Medication adjustment (includes PO) $\leq 3\text{d}$ ^(28, 29)
- **New onset infection actual or suspected and, Both:**
 - **Finding, $\geq$ One:**
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽³⁰⁾
 - **WBC, One:** ⁽³¹⁾
 - **Age ≤ 12 mos and, One:**
 - WBC $\leq 5,000/\text{cu.mm}(5 \times 10^9/\text{L})$
 - WBC $\geq 17,500/\text{cu.mm}(17.5 \times 10^9/\text{L})$
 - **Age > 1 to 5 yrs and, One:**
 - WBC $\leq 6,000/\text{cu.mm}(6 \times 10^9/\text{L})$
 - WBC $\geq 15,500/\text{cu.mm}(15.5 \times 10^9/\text{L})$
 - **Age > 5 to 12 yrs and, One:**
 - WBC $\leq 4,500/\text{cu.mm}(4.5 \times 10^9/\text{L})$
 - WBC $\geq 13,500/\text{cu.mm}(13.5 \times 10^9/\text{L})$
 - **Age > 12 to < 18 yrs and, One:**
 - WBC $\leq 4,500/\text{cu.mm}(4.5 \times 10^9/\text{L})$
 - WBC $\geq 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$
 - Bands $> 10\%(0.10)$
 - ANC $< 500/\text{cu.mm}(500 \times 10^6/\text{L})$ ⁽³²⁾
 - **Intervention, $\geq$ One:**
 - Culture pending $\leq 2\text{d}$
 - Drug-induced fever suspected and precipitating drug discontinued $\leq 3\text{d}$ ⁽³³⁾
 - Imaging study $\leq 24\text{h}$
- **Pain uncontrolled and, Both:**
 - Pain assessment ⁽³⁴⁾
 - Requiring change in medication, dose, or frequency ^(35, 36)
 - Post Critical care $\leq 24\text{h}$
- **INTERMEDIATE, $\geq$ One:**

- Post bone marrow or stem cell transplant and, **All:**
 - Awaiting engraftment
 - Isolation
- Hydration or nutrition, **≥ One:**
 - IV fluid ^(6, 7)
 - Advancing diet as tolerated or nutritional route established ⁽⁸⁾
- NIPPV ⁽¹⁷⁾
- Oxygen 35-39%(0.35-0.39) ⁽¹⁹⁾
- **CRITICAL, ≥ One:**
 - Hemodynamic instability, **Both:**
 - Finding, **≥ Two:**
 - Heart rate, **One:** ⁽¹⁰⁾
 - > 180/min, sustained (age ≤ 12 mos) ⁽¹¹⁾
 - > 160/min, sustained (age > 12 mos to 4 yrs) ⁽¹¹⁾
 - > 150/min, sustained (age > 4 yrs to 8 yrs) ⁽¹¹⁾
 - > 140/min, sustained (age > 8 yrs to 15 yrs) ⁽¹¹⁾
 - > 130/min, sustained (age > 15 yrs to < 18 yrs) ⁽¹¹⁾
 - Systolic BP, **One:**
 - < 70 mmHg (age ≤ 12 mos)
 - < 80 mmHg (age > 1 to < 5 yrs)
 - < 90 mmHg (age 5 to < 18 yrs)
 - Labile ⁽¹²⁾
 - MAP, **One:** ⁽¹³⁾
 - < 55 mmHg (≤ 12 mos)
 - < 65 mmHg (> 1 to < 18 yrs)
 - Intervention, **≥ One:**
 - Blood product transfusion ⁽¹⁴⁾
 - IV fluid resuscitation ≤ 2d ⁽¹⁵⁾
 - Hemodynamic monitoring, invasive ⁽¹⁶⁾
 - Mechanical ventilation
 - NIPPV and, **≥ One:** ⁽¹⁷⁾
 - FiO₂ ≥ 40%(0.40)
 - Respiratory intervention every 1-2h ⁽¹⁸⁾
 - Oxygen ≥ 40%(0.40) ⁽¹⁹⁾
 - Vasopressor, vasodilator, or inotrope continuous (excludes low or fixed dose for end stage heart failure)
 - **Post-procedure Day 31, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 24h, **All:** ⁽²⁰⁾
 - Afebrile
 - ANC ≥ 500/cu.mm(500x10⁶/L)
 - Platelet count > 20,000/cu.mm(20x10⁹/L)
 - Tolerating PO or nutritional route established ^(8, 21)
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽²²⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - Pain controlled or manageable ⁽²³⁾
 - GVHD resolving or stabilized
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **INTERMEDIATE, One:**

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**

- Use Extended Stay subset
- Refer for secondary review

● **CRITICAL, One:**

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**

- Use Extended Stay subset
- Refer for secondary review

Discharge, Level of Care, One: ⁽³⁷⁾

● **HOME, Both:**

● Level of care appropriateness, **Both:**

- Home environment safe and accessible ⁽³⁸⁾
- Patient or caregiver demonstrates ability to manage care

● Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁹⁾
- Medication reconciliation ⁽⁴⁰⁾
- Follow-up care planned ⁽⁴¹⁾
- Identify and address transportation needs

● **HOME CARE, Both:**

● Level of care appropriateness, **All:**

- Home environment safe and accessible ⁽³⁸⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(42, 43)

● Skilled services, ≥ **One:** ⁽⁴⁴⁾

- Skilled nursing ⁽⁴⁵⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽⁴⁶⁾

● Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁹⁾
- Medication reconciliation ⁽⁴⁰⁾
- Follow-up care planned ⁽⁴¹⁾
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**

● Level of care appropriateness, **One:**

- Support systems available ⁽⁴⁷⁾

● Skilled treatment, ≥ **One:**

- Clinical assessment ⁽⁴⁸⁾
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation

● **SKILLED MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

● Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁴⁵⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(49, 50)
- Functional impairments requiring at least supervision ⁽⁵¹⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁴⁰⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

● **Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 2x/wk$
- Treatment precluded in a lower level

● **Services, $\geq$ One:**

- Medical and skilled nursing interventions or assessment 4h/24h ⁽⁴⁵⁾

● **Therapy, All:**

- Goal directed therapy with rehabilitation potential ^(49, 50)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵¹⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 1 therapy discipline

● **Complete prior to facility transfer, All:**

- Medication reconciliation ⁽⁴⁰⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE REHAB, Both:**

● **Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- Treatment precluded in a lower level

● **Services, All:**

- Goal directed therapy with rehabilitation potential ^(49, 50)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵¹⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines

● **Complete prior to facility transfer, All:**

- Medication reconciliation ⁽⁴⁰⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● **Level of care appropriateness, Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$

● **Services, All:**

- Goal directed therapy with rehabilitation potential ^(49, 50)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵¹⁾
- Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁵²⁾

● **Complete prior to facility transfer, All:**

- Medication reconciliation ⁽⁴⁰⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

● **Level of care appropriateness, Both:**

- Treatment precluded in a lower level

● **In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ One:**

- Primary condition or illness and treatment of active comorbid condition(s) ⁽⁵³⁾
- Ventilator dependent $\geq 6h/d$ and weaning planned ^(54, 55)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽⁵⁶⁾

● **Complete prior to facility transfer, All:**

- Medication reconciliation ⁽⁴⁰⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Notes:**1:**

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

2:

The most common indications for bone marrow or stem cell transplants are leukemia, lymphoma, multiple myeloma, solid tumor cancers, or certain non-malignant conditions (e.g., congenital defects). Chemotherapy, radiation, and/or immunosuppressive drugs are used to prepare the patient for the transplant.

3:

The period preceding a bone marrow or stem cell transplant is used to prepare the patient for the transplant by completely suppressing the bone marrow and the patient's immune response. This conditioning regimen can consist of either or both chemotherapy and radiotherapy (Bazinet and Popradi, Curr Oncol 2019, 26: 187-91). The American Cancer Society suggests that 1-2 weeks is an appropriate time frame for patients to be admitted prior to a bone marrow or stem cell transplant. InterQual® external peer reviewers agree that 10 days captures the majority of the patients admitted for this treatment.

4:

Submyeloablative procedures are not included in this 7-day period unless the pre-transplant chemotherapy regimen requires administration in the inpatient setting or other co-morbidities necessitate acute inpatient care.

5:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

6:

To allow for the advancement of feedings as intravenous (IV) fluids are weaned, the IV fluid rates are $\geq 50\%$ (0.50) of the maintenance fluid requirement. These rates can be calculated using the following (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020):

Weight ≤ 10 kg = ≥ 50 mL/kg/24h

- Hourly rate = $2.08 \text{ (mL/h)} \times \text{kg}$

Weight > 10 -25 kg = ≥ 40 mL/kg/24h

- Hourly rate = $1.67 \text{ (mL/h)} \times \text{kg}$

Weight > 25 -60 kg = ≥ 30 mL/kg/24h

- Hourly rate = $1.25 \text{ (mL/h)} \times \text{kg}$

Example: An 11 kg child is required to receive a minimum of 40 mL/kg/24h to meet this criteria point. Multiply $1.67 \text{ (mL/h)} \times 11$ (child's weight in kg) = 18.37 mL/h. This is the minimum fluid replacement rate for this child.

7:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

8:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

9:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

10:

In the pediatric patient, hemodynamic instability could be assessed by an elevated heart rate (HR) above the 95th percentile norms based on specific age groups. A combination of different parameters and their trends could also be utilized to ascertain impending clinical deterioration more accurately in these patients to allow for closer monitoring in the critical care setting, including respiratory rate, mental status, and capillary refill time, amongst others. Other factors that could impact HR in the pediatric population include pain, anxiety, medications, and/or fever. Although no specific consensus has been reached on HR thresholds that signify unstable hemodynamics by various pediatric societies, systemic reviews and observational studies have compared HR data to textbook references such as the Pediatric Advance Life Support (PALS). Eytan et al., in 2017, studied HR in pediatric patients who were critically ill in the pediatric intensive care unit (PICU) and developed percentiles of heart rates categorized into the same age groups used in previous studies, including PALS, with the goal of comparing their observed norms, especially between the 5th and 95th percentiles (Eytan et al., *Frontiers in Pediatrics* 2017, 5). Using their HR data above the 95th percentile, HR thresholds were utilized to identify the pediatric patient who may require immediate or urgent intervention and rapidly treat unstable hemodynamics to prevent a potentially serious event.

11:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

12:

Blood pressure is considered labile when it is extremely variable and difficult to control.

13:

In pediatric patients, hemodynamic instability may be defined as a mean arterial pressure (MAP) less than 55 mmHg in infants up to 1 year of age and less than 65 mmHg in patients over 1 year of age (Matics and Sanchez-Pinto, *JAMA Pediatr* 2017, 171: e172352).

14:

The volume of blood product infused in a pediatric patient is dependent on the weight of the child. For some children, small volume transfusions of blood products (especially packed red blood cells) are necessary to prevent fluid overload. An appropriate transfusion volume would be 10 to 15 mL/kg, infused at a rate of 2 to 4 mL/kg per hour (Park, In: Elsevier, ed. *The Harriet Lane Handbook*, 23 edn. 2024:365-407; Rodrigue and Sloan, In: *Hematology*, 8 edn. 2023:1892-9).

15:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children:

Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

16:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous pressure, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

17:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

18:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

19:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air [room air: 21%(0.21) FIO₂] to correct hypoxia. For pediatric patients, the selection of an oxygen delivery system needs to take into consideration the patient's size, needs, and therapeutic goals. The oxygen concentration or percentage (FIO₂) delivered varies with the manufacturer's design, delivery device, patient's minute ventilation (dependent on the respiratory rate and tidal volume), and oxygen flow rate. In general, patients with higher minute ventilation

(common in most sick patients and those with respiratory disease) require considerably higher flow rates to correct hypoxia. The following are examples of oxygen delivery systems commonly used for pediatric patients with the percentage range of FIO_2 optimally delivered:

Low-flow systems

- Nasal cannula / Nasopharyngeal catheters 24-35%(0.24-0.35) FIO_2

Reservoir systems (For neonates, infants, and young children, the maximum FIO_2 achievable is not well documented)

- Simple oxygen masks 35-50%(0.35-0.50) FIO_2
- Partial-rebreathing masks 40-60%(0.40-0.60) FIO_2
- Non-rebreathing masks > 60%(0.60) FIO_2

High-flow systems

- Air-entrainment masks / nebulizers 24-40%(0.24-0.40) FIO_2
- Heated, humidified, high flow oxygen concentrating nasal cannula 21-100%(0.21-1.0) FIO_2

Enclosure systems

- Oxygen hoods / tents 21-100%(0.21-1.0) FIO_2

For children less than 5 years old, the oxygen regimen may need to be adjusted to a lesser concentration and flow rate, while still requiring acute level of care due to the patient's size, metabolic needs, and vital sign instability.

20:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

21:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (mL/kg/d) can be calculated based on current body weight (kg):

- 100 mL/kg/d for first 10 kg
- 50 mL/kg/d for second 10 kg
- 20 mL/kg/d for each additional kg in weight

22:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

23:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

24:

Graft versus Host Disease (GVHD) is a transplant complication that occurs when immunocompetent cells from a non-identical donor are grafted into an immunosuppressed recipient. The immunocompetent donor/graft cells recognize alloantigens in the host cells and the host develops a Graft vs Host (GVH) reaction. Classic acute GVHD typically presents during the first 100 days after a hematopoietic stem cell transplantation (HSCT). Classic chronic GVHD presents beyond the first 100 days post HSCT and may include overlap syndrome, with acute GVHD features. Classic acute GVHD affects the skin, gut, and liver. Diagnosis is strongly supported if symptoms include exanthema (an eruptive skin rash oftentimes associated with a fever or systemic symptoms), diarrhea, and elevated bilirubin levels. Chronic GVHD, in addition to skin, liver, or gut involvement, may also affect the mouth, eyes, respiratory tract, musculoskeletal system, and peripheral nerves. It presents with symptomatology resembling autoimmune diseases (e.g., systemic lupus, erythematosus, or rheumatoid arthritis).

Chronic GVHD grade can be categorized as mild, moderate, or severe. In patients with GVHD, hyperbilirubinemia and small intestinal/colonic involvement are known risk factors for increased mortality (Hill et al., Annu Rev Immunol 2021, 39: 19-49; Penack et al., Lancet Haematol 2020, 7: e157-e67; Rodrigues et al., American journal of clinical dermatology 2018, 19: 33-50).

In determining GVHD grade, stages for skin, liver and gut are scored.

Acute GVHD Staging and Grading (Rodrigues et al., American journal of clinical dermatology 2018, 19: 33-50):

Grade I: Stage 1-2 skin organ involvement, only

Grade II: Stage 3 skin; Stage 1 liver and gut

Grade III: Stage 1-3 skin; Stage 2-3 liver and Stage 2-4 gut

Grade IV: Stage 4 skin and liver

Stage 1

- Skin: Rash < 25%(0.25) of body surface area
- Liver: Bilirubin 2-3 mg/dL (35-40 µmol/L)
- Gut: Diarrhea > 500-1000 ml/day

Stage 2

- Skin: Rash 25%(0.25) to 50%(0.50) of body surface area with associated symptoms
- Liver: Bilirubin 3-6 mg/dL (53-102 µmol/L)
- Gut: Diarrhea > 1000 ml/day

Stage 3

- Skin: Generalized rash covering > 50%(0.50) of body surface area
- Liver: Bilirubin 6-15 mg/dL (104-225 µmol/L)
- Gut: Diarrhea > 1500 ml/day

Stage 4

- Skin: Generalized erythroderma (reddening of skin often associated with exfoliation) or bullous eruption
- Liver: Bilirubin > 15 mg/dL (225 µmol/L)
- Gut: Severe abdominal pain with or without ileus

Chronic GVHD grading (scores are based on a 4-point scale of 0-3):

- Mild: 1 or 2 organs, with score no more than 1 with lung score 0

- Moderate: 3 or more organs with no more than score 1 or > 1 organ (not lung) with score 2 or lung score 1
- Severe: At least 1 organ with score of 3 or lung score of 2 or 3

25:

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

26:

There have been significant improvements in immunosuppressant therapy for transplant recipients. Early protocols included azathioprine and corticosteroids, but with the addition of calcineurin inhibitors in solid organ transplantation, long-term survival and recovery has been improved. A multi-drug approach to therapy allows lower dosing of potentially very toxic individual agents. The 3 phases of immunosuppressive therapy include (Enderby and Keller, The American journal of managed care 2015, 21: s12-23):

- *Induction* - Immediately after transplant, when the risk for rejection is highest, intense antibody therapy or higher doses of maintenance therapy immunosuppressive agents are given depending on the organ being transplanted and the transplant center. Antibody therapies are T-cell depleting polyclonal (e.g., antithymocyte globulin (ATG)) and monoclonal (e.g., alemtuzumab or anti- CD52 antibody) or nondepleting agents (e.g., basiliximab).
- *Maintenance* - Transplant recipients remain on one or a combination of maintenance immunosuppressive therapy for the rest of their lives. Calcineurin inhibitors (e.g., cyclosporine, tacrolimus) are the mainstay of maintenance therapy with additional agents including antiproliferative/antimetabolites (e.g., mycophenolate mofetil (MMF), mycophenolic acid (MPA)), corticosteroids, mammalian target of rapamycin (mTOR) inhibitors (e.g., sirolimus), or T-cell costimulation blockers. The most common combination of these post transplant (kidney, liver, lung, heart, or pancreas) include tacrolimus, MMF/MPA, and prednisone. Immunosuppressant therapy may cause opportunistic infections, cancers and other adverse side effects. To limit these effects and increase survival and quality of life, research is underway that includes the infusion of marrow-derived autologous mesenchymal stem cells (MSCs), xenotransplantation, and advancements in regenerative medicine.
- *Rejection* - Anti-rejection therapy is given to reverse an episode of rejection and is based on the severity and type of rejection, as well as the hemodynamic consequences of rejection. Treatment includes immunosuppressant medications given for treating an acute rejection during the initial post-transplant and may consist of a medication adjustment, steroid administration, or the addition of T-cell depleting or nondepleting agents.

27:

Instruction: If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

28:

Medication adjustments tend to be driven by patient response. Patients being treated for symptoms of rejection usually have a biopsy and their medications adjusted based on the biopsy results to control the rejection.

29:

All medications commonly used to prevent rejection or opportunistic infections in the post-transplant period cause some degree of side effects. One of the important tasks of the transplant team is to titrate the dosage to minimize toxicity while maximizing benefit. Often this fine-tuning is empiric. If a patient has an intolerable side effect from a particular drug, an alternative medication or different combination or dosage of medications is attempted. Tacrolimus and cyclosporine (CSA) are immunosuppressive agents that inhibit calcineurin and are given to solid organ transplant recipients to prevent and treat allograft rejection. Tacrolimus is less nephrotoxic and may be more cost-effective than CSA. However, new-onset diabetes should be closely monitored during the medication period.

The dosing of CSA is more difficult to adjust appropriately in children, especially in younger/smaller children (less than 2 years). This is thought to be due to their shorter intestine, leading to less absorption of the drug, and also their faster metabolism. More time may be needed to adjust the dose and determine the appropriate medication management in this population than in older children or adults. The goal is to preserve the graft without seriously harming the patient.

30:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

31:

Reference for pediatric WBC range: (Dellinger et al., Crit Care Med 2013, 41: 580-637).

32:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than 500/cu. mm($500 \times 10^6/L$) are severely neutropenic and have a very limited immunologic response to infection.

To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) x WBC.

Examples:

ANC calculation using a percentage of total WBC count:

WBC count: 1,500/cu.mm($1.5 \times 10^9/L$)

Segs: 20%(0.20) of the WBCs

Bands: 2%(0.02) of the WBCs

$(20\% + 2\%(0.20 + 0.02)) \times 1,500/\text{cu.mm} (1.5 \times 10^9/L)$

$= 330/\text{cu.mm} (330 \times 10^6/L)$

Normal range: 1,500-8,000/cu.mm($1.5-8.0 \times 10^9/L$)

An ANC of less than 500/cu.mm($500 \times 10^6/L$) is considered severely neutropenic.

ANC Calculation using the above numbers as whole numbers:

WBC count: 1.5

Segs: 0.2

Bands: 0.02

$(0.2 + 0.02) \times 1.5 = 0.33$

Normal range: 1.5 to 8.0

An ANC of less than 0.5 is considered severely neutropenic.

33:

A drug-induced fever is a non-infectious febrile response following the administration of a drug that resolves when the drug is discontinued. The patient is often well-appearing and demonstrates resolution of the underlying disease process, despite the presence of a fever. A drug-induced fever should be suspected when the cause of the fever remains unknown, despite a full diagnostic work-up, or when the patient is receiving a drug frequently associated with fever. The most common agents triggering fever include anticonvulsants, anti-infectives, cardiac medications, allopurinol, and heparin.

The following drugs are also capable of inducing hyperthermia (Walter and Carraretto, Journal of the Intensive Care Society 2015):

- Serotonin-noradrenalin reuptake inhibitors
- Selective serotonin reuptake inhibitors
- Monoamine oxidase inhibitors
- Tricyclic antidepressants
- Bupropion
- Opioids

- Central nervous system stimulants
- Anticholinergics
- Antipsychotics
- Antispasmodics
- Sympathomimetic agents

Treatment involves discontinuing medications that are unnecessary, recently initiated, or highly associated with inducing fever. Resolution of the fever should occur within 48 to 72 hours of discontinuing the precipitating agent (Patel and Gallagher, Pharmacotherapy 2010, 30: 57-69; Eyer and Zilker, Crit Care 2007, 11: 236).

34:

Pain assessment includes quality, intensity, location, frequency, duration, and exacerbating or alleviating factors. Uncontrolled pain should be monitored, assessed, and documented. Appropriate medication adjustment and other interventions should be taken to reduce pain.

35:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

36:

For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
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- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

37:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

38:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity,

running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

39:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

40:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

41:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

42:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home

would be cumbersome and places the patient at risk

- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

43:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

44:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

45:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

46:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care,

or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

47:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

48:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

49:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

50:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

51:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment,

and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations

- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

52:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

53:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

54:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

55:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

56:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.